

Microwave Heating Cycle SOP (1x Cycle: 1 Bowl)

Heat one bowl of food per episode. Open the door, put the **vented cover** on the bowl, put the bowl in the middle of the turntable, and close the door. Press **START**, heat for 60 seconds, then press **STOP**. Add one 20 second extension only if no steam is visible. Open the door, take the bowl out to the **bowl spot**, take the cover off, and stir the food three slow circles with the spoon. Close the door at the end.

Only the **right gripper** goes inside the microwave. Never open the door while the microwave runs.

The table is set up in one of three ways. Only the bowl moves; the microwave, the bowl spot, the cover spot, the spoon rest, and the spill cleanup spot are in the same place in all three.

- **Config L:** the bowl starts at the front-left, in front of the cover spot.
- **Config M:** the bowl starts on the bowl spot at center front.
- **Config R:** the bowl starts at the front-right, in front of the spoon rest.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the bowl starts at the front-left (**Config L**), on the bowl spot (**Config M**) or at the front-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the bowl's side brings it to the bowl spot — the **left gripper** in Config L, the **right gripper** in Config R. No arm reaches across the table.
- **Working position:** the bowl stands on the bowl spot before the cover goes on, and it comes back to the bowl spot after heating. Nothing is done to the bowl in the start zone except that first carry.
- **Fixed roles:** from the bowl spot on, every step is the same in all three configs. Only the right gripper presses the buttons, carries the bowl into and out of the microwave, and stirs; the left gripper works the door and the cover and holds the bowl for the stir.

Setup

Complete both checklists before starting the episode.

Cell configuration

- **Environment camera:** 900 mm.

- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows both table edges, the microwave and its control panel, the open door path, the bowl spot and the two other bowl start zones, the cover spot, the spoon rest, and the spill cleanup spot.
3. Both arms are at the home position with grippers open.
4. The microwave is plugged in, off, empty, and dry. Its door is closed and latched.
5. The turntable is seated flat and turns freely. The inside light works.
6. The table is clean, dry, and clear except for the task materials.

Materials checklist

1. One bowl of food stands upright in the start zone for this episode's config. The other two zones are empty.
 - **Config L:** front-left, in front of the cover spot
 - **Config M:** on the bowl spot at center front
 - **Config R:** front-right, in front of the spoon rest
2. The bowl is rigid, safe for the microwave, dry on the outside, and has no metal on it.
3. The bowl is filled no more than halfway, so food does not spill during a level carry.
4. The bowl has a wide flat rim the right gripper can hold.
5. One clean, dry **vented cover** lies on the cover spot on the left. It is safe for the microwave and sits flat on the bowl rim without being pushed down.
6. One stirring spoon lies on the spoon rest on the right. It is plastic or wood, not metal, and it never goes inside the microwave.
7. The microwave has one visible **DOOR RELEASE** button, one **START** button, and one **STOP** button, all on the control panel to the right of the door.
8. One cloth and one waste tray are staged at the front spill cleanup spot. Spilled food is never put back in the bowl.

Workspace layout

- **Center back:** the microwave. The door is hinged on the left and swings open to the left. The control panel is to the right of the door.
- **Center front:** the bowl spot. The bowl ends here in every config, and starts here in Config M.
- **Bowl start zone:** front-left, in front of the cover spot (**Config L**); the bowl spot (**Config M**); or front-right, in front of the spoon rest (**Config R**). One per episode.

- **Left:** the cover spot with the vented cover.
- **Right:** the spoon rest with the stirring spoon.
- **Front:** spill cleanup spot with the cloth and the waste tray.

Arm assignments

- **Right gripper:** all the main work. It presses the three buttons, carries the bowl in and out, and stirs with the spoon. It is the only gripper that goes inside the microwave. In Config R it also brings the bowl from the front-right to the bowl spot.
- **Left gripper:** the supporting work. It opens and closes the door, moves the cover, and holds the bowl steady during the stir. It never goes inside the microwave. In Config L it also brings the bowl from the front-left to the bowl spot.

Vocabulary

- **Start zone:** where the bowl stands at the start of the episode — front-left (**Config L**), the bowl spot at center front (**Config M**), or front-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Open door:** the door is released by the **DOOR RELEASE** button and swung fully to the left. It stays open by itself.
- **Latched door:** the door is pushed shut and the door edge is flat against the microwave front.
- **Vented cover:** the loose lid that sits on the bowl rim. It lets steam out, so it is never pressed down or clipped on.
- **Level carry:** the bowl stays flat while it moves, and the food surface does not tilt.
- **Centered bowl:** the bowl sits flat on the turntable with about the same gap all the way around it. Nothing touches the walls or the door.
- **Cycle running:** the inside light is on and the turntable is turning.
- **Cycle stopped:** the inside light is off and the turntable is not turning.
- **Steam cue:** you can see mist on the door window or steam under the cover.
- **Slow circle:** the spoon goes once all the way around the inside of the bowl, taking about one second, without lifting out.
- **Abnormal heating condition:** sparks, smoke, a loud pop, the cover lifting off, food boiling over, or the bowl or turntable shifting.

Steps

Heat the bowl once. Add one 20 second extension only when the check in Step 5 asks for it. Only Step 2 depends on where the bowl starts: in Config L the **left gripper** brings the bowl from the front-left to the bowl spot, in Config R the **right gripper** brings it from the front-right, and in Config M it already stands there. Every other line is the same in all three configs.

Step 1: Open the microwave door

Goal: the door is fully open before the bowl is loaded.

- With the **right gripper**, press and release the **DOOR RELEASE** button, then move clear.
- With the **left gripper**, hook the front edge of the door and swing it fully open to the left. Release it and move the left gripper clear.

Check: the door stays fully open by itself and the inside of the microwave is empty and dry. If the door does not pop open, press **DOOR RELEASE** once more with the **right gripper**. Do not pull a latched door open.

Step 2: Put the cover on the bowl

Goal: the bowl stands on the bowl spot and the vented cover sits flat on its rim.

Look where the bowl is before reaching for it.

- **IF the bowl is at the front-left (Config L):** with the **left gripper**, hold the bowl by its rim on the left side, lift it just clear of the table, and carry it level to the right to the bowl spot. Lower it until the base rests flat, then release and move clear.
- **IF the bowl is on the bowl spot (Config M):** the bowl is already at the working position. Do not move it.
- **IF the bowl is at the front-right (Config R):** with the **right gripper**, hold the bowl by its rim on the right side, lift it just clear of the table, and carry it level to the left to the bowl spot. Lower it until the base rests flat, then release and move clear.

Then, in all three:

- With the **right gripper**, hold the bowl rim to keep it still.
- With the **left gripper**, pick up the cover from the cover spot and hold it level.
- With the **left gripper**, lower the cover straight down onto the bowl rim, then release and move clear.
- Release the bowl with the **right gripper**.

Check: the bowl stands upright on the bowl spot, and the cover sits flat on the rim and does not rock. If it sits crooked, lift it straight up with the **left gripper** and lower it again. Do not press the cover down.

Step 3: Put the bowl in the microwave

Goal: the covered bowl sits flat in the middle of the turntable.

3.1 Lift the bowl

- With the **right gripper**, hold the bowl by its rim on the right side.
- Lift the bowl just clear of the table and keep it level.

3.2 Set the bowl on the turntable

- With the **right gripper**, carry the bowl level through the door opening.
- Lower it until it rests flat in the middle of the turntable, then release and pull the right gripper straight back out.
- Keep the **left gripper** outside the microwave and clear of the door.

Check: the bowl is centered, sits flat, and the cover is still on. Nothing touches the walls or the door. If the bowl is not centered, take it out with the **right gripper**, then set it down again. Do not push it across the turntable.

Step 4: Close the door

Goal: the door is latched before heating.

- With the **left gripper**, push the front edge of the door shut until it latches, then release and move clear.
- Move both grippers away from the door and the control panel.

Check: the door edge is flat against the microwave front. If it did not latch, push it once more with the **left gripper**. Never start the microwave with the door open.

Step 5: Heat for 60 seconds and extend once if needed

Goal: heat for 60 seconds, and add 20 more seconds only if no steam is visible.

5.1 Start the cycle

- Make sure both grippers are clear of the door and the control panel.
- With the **right gripper**, press and release **START**, then move clear.

5.2 Wait and stop

- Wait 60 seconds. Do not touch the microwave, the door, or the table while the cycle runs.
- After 60 seconds, press and release **STOP** with the **right gripper**, then move clear.
- Wait until the inside light is off and the turntable is not turning.

5.3 Look for steam and extend once if needed

- Look through the door window. Do not open the door and do not touch the microwave.
- If you see the steam cue, go to Step 6.
- If you see no steam, press and release **START** with the **right gripper**, then move clear.
- Wait 20 seconds without touching anything. Then press and release **STOP** with the **right gripper** and wait until the cycle is stopped.
- Look again. Never run a third cycle. If you still see no steam, do not open the door. End the episode unsuccessful and check the food and the microwave with recording off.

Abnormal condition branch: press and release **STOP** at once with the **right gripper**, move both arms clear, and leave the door closed. Do not start again and do not open the door. End the episode unsuccessful and check the microwave with recording off.

Check: the microwave ran 60 seconds, plus 20 seconds only when the steam check asked for it. The cycle is stopped. If a cycle stopped early on its own and nothing was abnormal, do not start it again; tag the violation and end the episode once the cycle is stopped.

Step 6: Open the door after the cycle

Goal: open the door only after the cycle is stopped.

- Make sure the inside light is off and the turntable is not turning.
- With the **right gripper**, press and release **DOOR RELEASE**, then move clear.
- With the **left gripper**, swing the door fully open to the left, then release and move clear.

Check: the door stays fully open and the bowl is still upright and centered. If the door does not pop open, press **DOOR RELEASE** once more with the **right gripper**.

Step 7: Take the bowl out

Goal: the covered bowl stands upright on the bowl spot.

7.1 Lift the bowl out

- With the **right gripper**, hold the bowl by its rim on the right side.
- Lift it straight up just clear of the turntable, keep it level, and bring it straight out through the door opening.
- Keep the **left gripper** clear of the door and outside the microwave.

7.2 Set the bowl down

- With the **right gripper**, carry the bowl level to the bowl spot at center front.
- Lower it until the base rests flat, then release.

Check: the bowl is upright and steady on the bowl spot, the cover is still on, and no food has spilled. Do not slide the bowl across the turntable edge and do not tilt it.

Step 8: Take the cover off

Goal: the cover is off the bowl and back on the cover spot.

- With the **right gripper**, hold the bowl rim to keep it still.
- With the **left gripper**, lift the cover straight up and keep it level, so drips fall back into the bowl.
- With the **left gripper**, carry the cover to the cover spot and lay it down flat. Release it and move clear.

- Release the bowl with the **right gripper**.

Check: the bowl is open and steady, and the cover lies flat on the cover spot. Do not tip the cover over the table and do not drop it beside its spot.

Step 9: Stir the food

Goal: the food is stirred three slow circles and the spoon is back on its rest.

9.1 Pick up the spoon and hold the bowl

- With the **left gripper**, hold the bowl rim to keep it still.
- With the **right gripper**, pick up the spoon by its handle from the spoon rest.

9.2 Stir and return the spoon

- With the **right gripper**, lower the spoon into the food until it reaches the bottom of the bowl.
- Make three slow circles around the inside of the bowl. Keep the spoon under the food surface and do not splash.
- Lift the spoon straight out and let it drip over the bowl for 2 seconds.
- With the **right gripper**, lay the spoon back on the spoon rest and release it.
- Release the bowl with the **left gripper**.

Check: the food looks mixed, nothing splashed out, and the spoon lies on its rest. If the spoon was lifted out early, put it back in with the **right gripper** and finish the three circles. Do not bang the spoon on the rim.

Step 10: Close the door, clean spills, and end the episode

Goal: the door is latched, spills are cleaned, and both arms are home.

10.1 Close the door

- With the **left gripper**, push the front edge of the door shut until it latches, then release and move clear.

Check: the door edge is flat against the microwave front. If it did not latch, push it once more with the **left gripper**.

10.2 Clean spills

- If you see no spill, go to Step 10.3.
- With the **right gripper**, pick up the cloth by one corner and wipe spills inward, away from the control panel and the bowl.
- Push solid bits into the waste tray. Keep wiping until the table and the bowl base are clear.
- Never put spilled food back in the bowl.
- With the **right gripper**, put the cloth back on the cleanup spot.

- If liquid reached the control panel, do not touch it. Move both arms clear, end the episode, then unplug and clean the microwave with recording off.

10.3 End the episode

- Confirm the bowl is upright and open on the bowl spot, the cover lies on the cover spot, the spoon lies on its rest, the door is latched, and no spill is left.
- Return both arms home with grippers open, then end data collection.

After the episode: reset the workspace

These actions happen while recording is stopped.

- Take the heated bowl off the table and empty it under the station procedure.
- Wash and dry the bowl, the cover, and the spoon.
- Wipe and dry the turntable, the inside of the microwave, the door, and the table.
- Check that the turntable is seated flat and turns freely, and that the microwave is off, empty, and dry with its door latched.
- Stage a fresh bowl of food, filled no more than halfway, upright in the start zone for the next episode's config — front-left (Config L), the bowl spot (Config M), or front-right (Config R). Put the clean cover on the cover spot and the clean spoon on the spoon rest.
- Empty the waste tray and swap the cloth if it was used.
- Run both Setup checklists before the next recording.

SOP violations

Actions that break the SOP and are scored in side by side review.

How to record a violation in review

For each violation spotted in a recorded episode, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken**.

The visible cue is seen in the video. The coaching note supports retraining and is not a label.

Episode handling

Tag each violation with its timestamp and name. Retain the tagged episode in training data.

Violations

Note on the start position: the violations below were written for Config M (the bowl starts on the bowl spot at center front). The pickup and arm-role cues will be rewritten later to cover all

three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the bowl is not in the start zone for the config; a gripper reaches across the table for the bowl; the cover goes on or the bowl is lifted toward the microwave anywhere but on the bowl spot; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the left gripper brings the bowl to the bowl spot in Config L, the right gripper in Config R, and in Config M the bowl is not moved before the cover goes on; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the bowl is before the first reach, then follow that config's IF line in Step 2.

Violation: Door not fully opened before loading

- **Visible cue:** the bowl is moved toward the microwave while the door is shut or only part way open, or the door is pulled open without pressing DOOR RELEASE.
- **SOP rule broken:** Step 1 (press DOOR RELEASE with the right gripper, swing the door fully open with the left gripper, and confirm it stays open before loading).
- **Coaching note:** open the door and check it before you pick up the bowl.

Violation: Cover not placed correctly

- **Visible cue:** the cover sits crooked or rocks on the rim, it is pressed or clipped down, or the bowl goes into the microwave with no cover on it.
- **SOP rule broken:** Step 2 (lower the cover straight down with the left gripper so it sits flat on the rim, and never press it down).
- **Coaching note:** lower the cover flat and leave it loose.

Violation: Bowl not held steady while the cover is moved

- **Visible cue:** the bowl slides or rocks while the cover is put on or taken off because the other gripper is not on the rim.
- **SOP rule broken:** Steps 2 and 8 (hold the bowl rim with the right gripper while the left gripper moves the cover).
- **Coaching note:** hold the bowl first, then move the cover.

Violation: Bowl not carried level

- **Visible cue:** the bowl tilts, swings, or is carried high during any carry, or food sloshes over the rim.

- **SOP rule broken:** Steps 3 and 7 (lift the bowl just clear and keep it level through the whole carry).
- **Coaching note:** lift a little, keep it flat, move slowly.

Violation: Bowl not centered on the turntable

- **Visible cue:** the bowl sits near the edge, touches a wall or the door, or does not sit flat.
- **SOP rule broken:** Step 3 (set the bowl flat in the middle of the turntable, with about the same gap all the way around).
- **Coaching note:** aim for the middle before you release.

Violation: Bowl pushed across the turntable

- **Visible cue:** a bowl that is not centered is dragged or nudged into place instead of being taken out and set down again.
- **SOP rule broken:** Step 3 (fix a bowl that is not centered by taking it out with the right gripper and setting it down again, not by pushing it).
- **Coaching note:** take it out and place it again.

Violation: Left gripper enters the microwave

- **Visible cue:** the left gripper or left arm crosses the door opening at any point.
- **SOP rule broken:** Steps 3, 6 and 7 (only the right gripper goes inside the microwave; the left gripper stays outside and clear of the door).
- **Coaching note:** the left arm owns the door only.

Violation: Door not latched before starting

- **Visible cue:** START is pressed while the door is open or resting unlatched against the front.
- **SOP rule broken:** Step 4 (push the door shut with the left gripper and confirm it is latched before heating).
- **Coaching note:** push the door until it latches, then look at the edge.

Violation: Gripper not clear when a button is pressed

- **Visible cue:** the other gripper is still on the door, on the bowl, or over the control panel when DOOR RELEASE, START, or STOP is pressed.
- **SOP rule broken:** Steps 4, 5.1 and 5.2 (move both grippers clear of the door and the control panel before pressing a button, and move clear after each press).
- **Coaching note:** clear both arms first, then press.

Violation: Incorrect heating cycle

- **Visible cue:** the first cycle runs shorter or longer than 60 seconds; the extension is not 20 seconds; an extension is run when steam was already visible or skipped when it was not; a

third cycle starts; or a cycle that stopped early is started again.

- **SOP rule broken:** Step 5 (run 60 seconds, add one 20 second extension only when no steam is visible, never run a third cycle, and never restart a cycle that stopped early).
- **Coaching note:** count 60, stop, look, and add 20 only if you must.

Violation: Steam check skipped or not confirmed

- **Visible cue:** the extension decision is made without looking through the door window, or the door is opened to check instead.
- **SOP rule broken:** Step 5.3 (look through the door window with the door closed before deciding on the extension).
- **Coaching note:** look through the window every time; the door stays shut.

Violation: Microwave touched while the cycle runs

- **Visible cue:** a gripper touches the door, the control panel, or the microwave body while the inside light is on and the turntable is turning.
- **SOP rule broken:** Step 5.2 (do not touch the microwave, the door, or the table while the cycle runs).
- **Coaching note:** hands off until the cycle is stopped.

Violation: Door opened before the cycle stopped

- **Visible cue:** DOOR RELEASE is pressed or the door is swung open while the inside light is on or the turntable is still turning.
- **SOP rule broken:** Steps 5.2 and 6 (wait until the light is off and the turntable is not turning before opening the door).
- **Coaching note:** wait for the light and the turntable, then open.

Violation: Abnormal heating condition not handled correctly

- **Visible cue:** the cycle keeps running after sparks, smoke, a loud pop, the cover lifting, food boiling over, or the bowl or turntable shifting; or the cycle is started again or the door is opened afterward.
- **SOP rule broken:** Step 5 abnormal condition branch (press STOP at once, clear both arms, leave the door closed, do not start again, and end the episode unsuccessful).
- **Coaching note:** stop at the first bad sign and leave the door shut.

Violation: Food heated after a failed steam check

- **Visible cue:** the door is opened or the bowl is taken out after the extension while no steam is visible.
- **SOP rule broken:** Step 5.3 (if no steam is visible after the extension, do not open the door; end the episode unsuccessful).

- **Coaching note:** no steam after the extension means stop, not continue.

Violation: Cover removed incorrectly

- **Visible cue:** the cover is tipped over the table, dripped outside the bowl, dropped, or left somewhere other than flat on the cover spot.
- **SOP rule broken:** Step 8 (lift the cover straight up and level with the left gripper, then lay it flat on the cover spot).
- **Coaching note:** keep the cover level until it is over its spot.

Violation: Incorrect stir

- **Visible cue:** fewer or more than three circles, a fast stir, the spoon lifted out before the three circles are done, the spoon not reaching the bottom, food splashed out, or the spoon banged on the rim.
- **SOP rule broken:** Step 9.2 (lower the spoon to the bottom and make three slow circles under the food surface without splashing or banging).
- **Coaching note:** three slow circles, spoon down, no splash.

Violation: Bowl not held steady during the stir

- **Visible cue:** the bowl slides or spins on the bowl spot while the spoon is in the food because the left gripper is not on the rim.
- **SOP rule broken:** Step 9.1 (hold the bowl rim with the left gripper before the spoon goes in).
- **Coaching note:** hold the bowl first, then stir.

Violation: Spoon not returned to its rest

- **Visible cue:** the spoon is left in the bowl, laid on the table, or dropped beside the rest.
- **SOP rule broken:** Step 9.2 (lay the spoon back on the spoon rest and release it).
- **Coaching note:** the spoon goes back on the rest every time.

Violation: Door left open or unlatched at the end

- **Visible cue:** the door is open or resting unlatched when the episode ends.
- **SOP rule broken:** Step 10.1 (push the door shut with the left gripper until it latches).
- **Coaching note:** close the door before you finish.

Violation: Spill handled incorrectly

- **Visible cue:** spilled food is put back in the bowl, the wet control panel is touched, or a spill is still there when the episode ends.
- **SOP rule broken:** Steps 10.2 and 10.3 (wipe spills into the waste tray, never put spilled food back, never touch a wet control panel, and confirm the table is clear before ending).

- **Coaching note:** clean after the door is shut, and throw spilled food away.

Violation: Step skipped

- **Visible cue:** a step is never done — the cover is never put on, the bowl is heated without being centered, the stir never happens, the cover is never taken off, or a **Check** is skipped and the next step starts anyway.
- **SOP rule broken:** Steps 1 to 10 (do every step in order and finish it, including its **Check**, before starting the next).
- **Coaching note:** say the next step before you move; a step whose check you never made is not done.

Violation: Wrong arm used

- **Visible cue:** a motion given to one gripper is done by the other — the right gripper works the door, the left gripper presses a button or carries the bowl, or the left gripper stirs.
- **SOP rule broken:** Steps 1 to 10 (each action uses the gripper its step names).
- **Coaching note:** left arm on the door and the cover, right arm on the buttons, the bowl, and the spoon.

Violation: Object dropped or knocked over

- **Visible cue:** the bowl, cover, or spoon is dropped, tipped, or knocked off its spot at any point.
- **SOP rule broken:** Steps 2, 3, 7, 8 and 9 (hold each item until it rests flat on its spot, then release).
- **Coaching note:** set it down fully before you let go.

Violation: Wrong episode ending

- **Visible cue:** recording ends before the bowl is upright and open on the bowl spot, the cover is on the cover spot, the spoon is on its rest, the door is latched, spills are cleaned, or both arms are home. Or an arm moves again after being sent home.
- **SOP rule broken:** Step 10 (close the door, clean spills, confirm the end state, home both arms, then end data collection).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

Log these failures as system issues, discard the episode, and do not coach on them.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, or motor error.

- **Defective microwave:** no power, a button that does not respond, a door that will not latch, a turntable that will not turn, or a cycle that stops on its own.
- **Defective material:** a cracked bowl, a warped cover that will not sit flat, a broken spoon, or food that does not steam after the 60 second cycle and the 20 second extension.

Annotation subtasks (from SOP)

1. Open the microwave door with DOOR RELEASE
2. Put the vented cover on the bowl
3. Put the covered bowl in the middle of the turntable
4. Close and latch the door
5. Run the 60 second heating cycle and, if needed, one 20 second extension
6. Open the door after the cycle stops
7. Take the bowl out to the bowl spot
8. Take the cover off and return it to the cover spot
9. Stir the food three slow circles and return the spoon
10. Close the door, clean spills, home both arms, and end the episode